

PRESENTED BY

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MICHELIN

Marketing Analytics Report

RESEARCH QUESTION



How do restaurants with different Michelin star levels (Selected, Bib, 1, 2, and 3) differ across key features (e.g. food, service, ambience)?

In this report, we explore how the emphasis on these features shifts across the Michelin ladder: whether higher-rated restaurants are characterized by systematically different patterns in food, service, cleanliness, ambience, and accessibility, and how different tiers appear to trade off “culinary mastery” against factors like atmosphere, value, and convenience within the Michelin system.

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I. Limitations

DATA OVERVIEW

Basic Structure

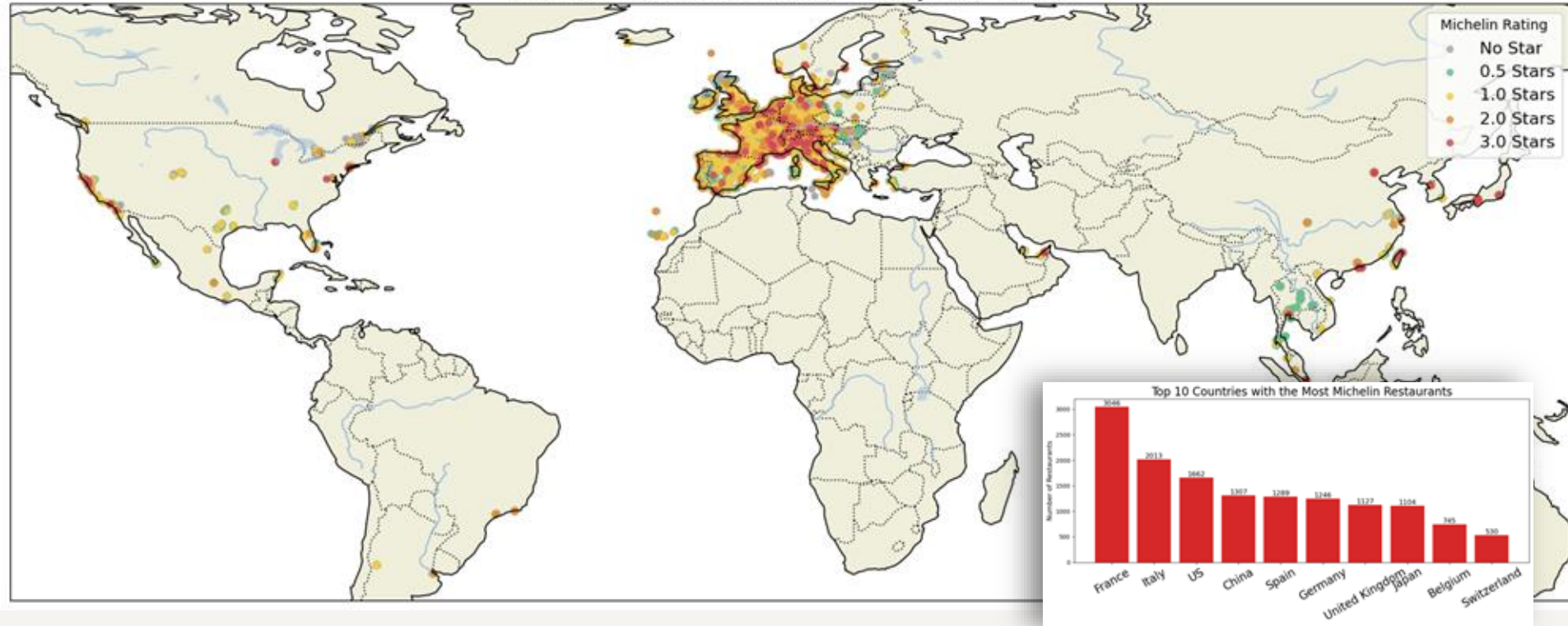
- **18,422** restaurants in total
- Representing **19** main countries
- Containing **1,711** unique cuisine labels



Column Name	Meaning / Description
Name	Name of the Michelin-listed restaurant.
Location	Combined City + Country text formatted by Michelin.
Country	Extracted country name.
Cuisine	Type of cuisine served (e.g., French, Japanese, Italian).
Award	Michelin distinction (Bib Gourmand, 1–3 stars).
Price	Price category assigned by Michelin (e.g., \$\$, \$\$\$).
GreenStar	Geolocation of the restaurant, used for mapping.
Description	Text description used for review/attribute extraction.
StarNumeric	Binary/count indicator of whether a specific attribute family is mentioned.
is_home_cuisine	1 if the cuisine matches the country of origin, else 0.
attr_*	Attribute-family level signals (e.g., FOOD, BEVERAGE, SERVICE).
feat_*	Fine-grained feature-level attributes. Each column = 1/0 indicating the specific detail mentioned (e.g., ingredient freshness, plating, staff friendliness).

VISUALIZATION - The World Michelin Map

Michelin Restaurant Locations by Star Level



API

- We used an LLM-based API to extract attributes and features from Michelin inspectors' descriptions.
- To discover recurring patterns, we sampled **10 random batches of 500 descriptions each** and extracted candidate features.
- These raw outputs were **manually consolidated into high-level attributes and their sub-features**.
- We then used the API to label each restaurant description with **binary indicators** (1 = mentioned, 0 = not mentioned).
- Sentiment scoring was tested on the first ~100 descriptions but returned uniformly high values, so we adopted **binary mention detection instead of sentiment analysis**.

FOOD QUALITY

- Ingredient freshness, quality & sourcing
- Seasonal, local & market-driven ingredients
- ...

BEVERAGE QUALITY

- Wine list breadth, curation & recommendations
- Cocktail craftsmanship & mixology quality
- ...

SERVICE QUALITY

- Staff friendliness, warmth & hospitality
- Professionalism, expertise & explanations
- ...

MENU VARIETY & CUSTOMIZATION

- Seasonal & rotating menus
- Menu formats (tasting menus, prix fixe, à la carte, sharing)
- ...

AMBIENCE & INTERIOR ENVIRONMENT

- Interior decor, aesthetic design & thematic style
- Lighting, music, noise & overall atmosphere
- ...

CLEANLINESS & HYGIENE

- Dining area cleanliness & upkeep
- Kitchen & food-prep hygiene standards
- ...

PRICE & VALUE

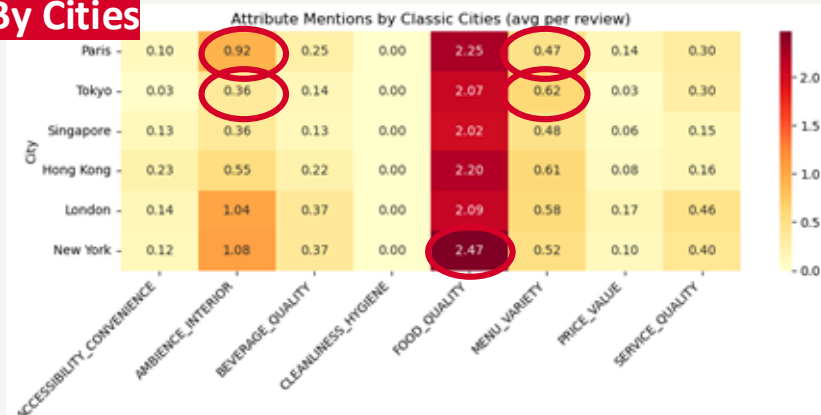
- Overall value for money
- Price level & market positioning
- ...

ACCESSIBILITY & CONVENIENCE

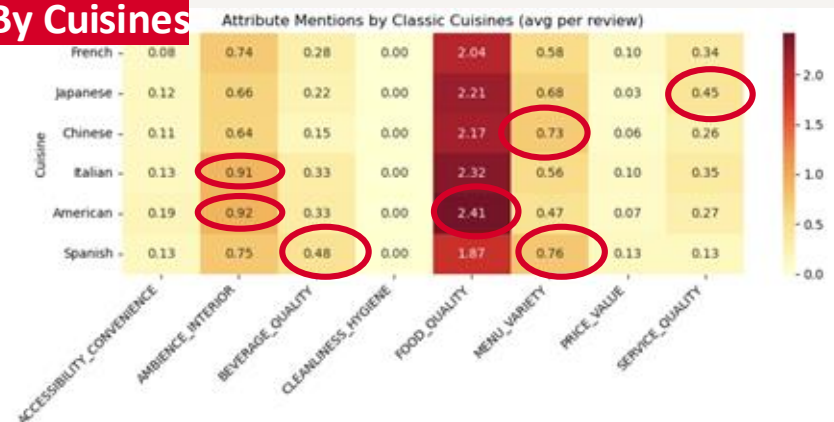
- Location convenience & transport access
- Parking availability & guest access infrastructure
- ...

VISUALIZATION - Description

By Cities



By Cuisines



Attribute Category	Count
Food Quality	17,824
Ambience & Environment	12,151
Menu Variety & Customization	8,547
Service Quality	5,174
Beverage Quality	4,852
Accessibility & Convenience	2,685
Price & Value	1,681
Cleanliness & Hygiene	51

1. Overall Insights

- **Food Quality, Ambience:** top drivers
- **Menu creativity** plays an important role
- **Price, location, cleanliness:** expected standards rather than differentiators in high-end dining.

2. City-Level Differences

Paris vs. Tokyo: Two Distinct Models of Fine Dining

- Paris: “Atmosphere-first.”
- Tokyo: “Craft-first.”
- **New York: The Most Food-Driven Market**
- Fine dining has different meaning across cultures.

3. Cuisine-Level Differences

Universally Food-first, but differ in their secondary priorities

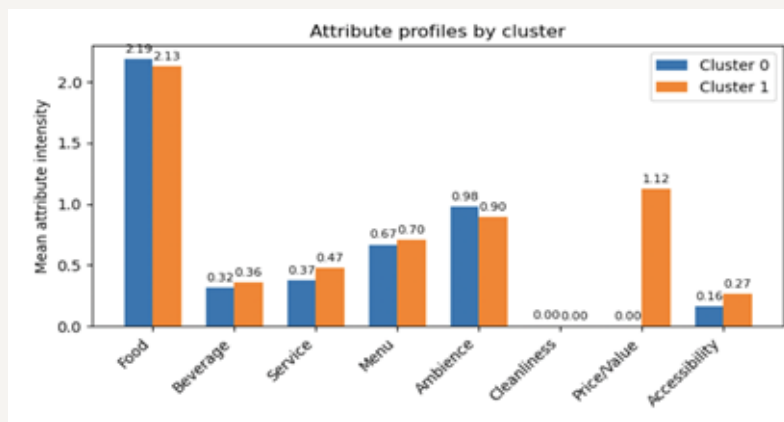
Different second-highest expectation:

- Japanese → Menu structure, omakase pacing
- Chinese → Menu variety (0.73, highest of all)
- Italian → Ambience-driven
- American → Ambience over menu diversity
- Spanish → Variety (0.76, highest)

CLUSTER ANALYSIS

- Clustering restaurants by text-based attribute profiles

- **Inputs:** 8 attribute-intensity scores per restaurant (*Food, Beverage, Service, Menu, Ambience, Cleanliness, Price/Value, Accessibility*).
- **Pre-processing:** standardised each attribute (z-scores).
- **Algorithm:** k-means on attribute space; tested $k = 2-6$.
- **Model choice:** $k = 2$ selected based on highest silhouette score (0.324) and clear interpretability.



Cluster 0 – “Ambience-led experiences”

- Cluster 0 restaurants are described in terms of **food and atmosphere**, with value and logistics largely taken for granted. Reviews frame these venues as immersive dining experiences rather than deals or practical choices.

Cluster 1 – “Value & practicality framed”

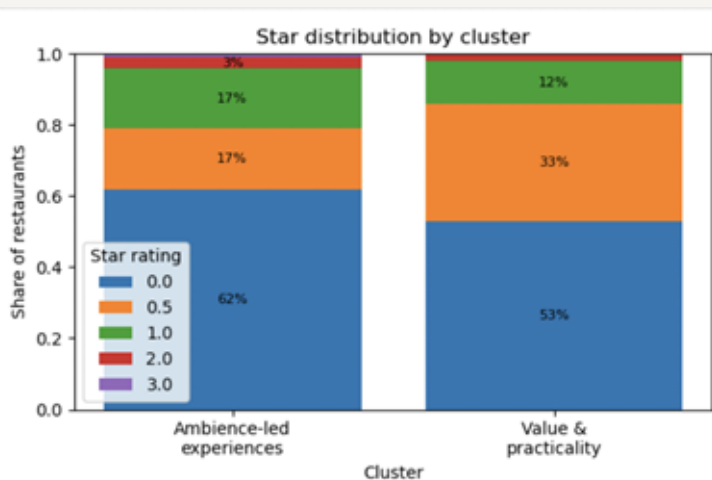
- Cluster 1 restaurant are characterised by texts that explicitly emphasise **affordability, fairness and ease of access**, on top of solid food and service. It represents a more **value- and practicality-oriented positioning** in the text.

CLUSTER ANALYSIS

- Cluster differences in Michelin outcomes & positioning

Star distributions across clusters are very similar → different narratives, similar recognition.

- Both clusters are dominated by 0 and 0.5 (Bib) restaurants; the shares of 1–3 stars are close.
- Cluster 1 has more **Bib Gourmand** restaurants (33% vs. 17%) but fewer fully starred restaurants overall; Cluster 0 has slightly more 1–3 star venues.



Positioning, not level

- The clusters mainly capture **how the experience is framed in text**(ambience-led vs. value-led), rather than radically different levels of Michelin success.

For restaurants

- Regardless of which “lane” a restaurant plays in — ambience-led or value-led — **moving up the Michelin ladder still depends on exceptional food and service**, not just on ambience or value.

PCA ANALYSIS

PC1: Practical Value → Experiential Offer → Culinary Mastery

Positive loadings:

- MENU (0.411)
- SERVICE (0.406)
- BEVERAGE (0.396)
- FOOD (0.352)
- AMBIENCE (0.276)

Negative loadings:

- PRICE (-0.349)
- ACCESSIBILITY (-0.380)

PC1 represents a continuum from functional dining propositions toward curated, ritualized culinary experiences.

PC2: Hospitality Narrative → Technical Precision

Negative loadings:

Positive loadings:

- CLEANLINESS (0.597)
- FOOD craft/quality (0.319)
- AMBIENCE (-0.500)
- ACCESSIBILITY (-0.318)
- PRICE (-0.287)

PC2 captures a shift from emotive, ambience-driven presentation toward operational discipline and culinary precision.

PC1 is the dominant dimension — it explains **nearly 70%** of variance.

=== Explained Variance ===

	ExplainedVariance
0	0.685195
1	0.255509

=== Star PCA Coordinates ===

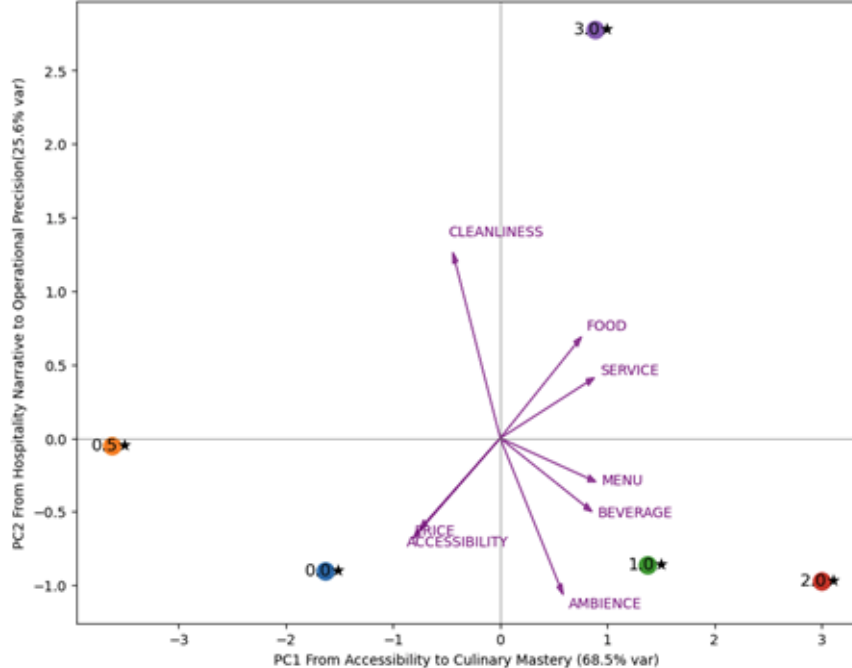
	PC1	PC2
StarNumeric		
0.0	-1.629876	-0.899964
0.5	-3.621279	-0.048761
1.0	1.376086	-0.860644
2.0	2.994877	-0.970206
3.0	0.880193	2.779574

=== PCA Attribute Loadings ===

	PC1_loading	PC2_loading
FOOD	0.351972	0.319700
BEVERAGE	0.396303	-0.231063
SERVICE	0.405770	0.189704
MENU	0.411075	-0.136510
AMBIENCE	0.275837	-0.499904
CLEANLINESS	-0.208984	0.596917
PRICE	-0.348766	-0.286793
ACCESSIBILITY	-0.379512	-0.318302

PERCEPTUAL MAP

Michelin Perceptual Map by Star Category (Attribute-level)



0-Star Restaurants: Practical, accessible, narrative-first

- Their stories lean on price fairness, approachability and atmosphere

0.5-Star: The value-performance sweet spot

- These restaurants **win on value, comfort, and reliability**, not culinary excellence.

1-Star Restaurants: Culinary mastery begins.

- Core value is **food quality and menu clarity**

2-Star Restaurants: Craft, Curation, Intentionality

- They peak in menu innovation and storytelling through food.

3-Star Restaurants: Ritualized perfection

- They no longer “prove their cuisine.” they manage precision, environment, time, attention and emotion.

Michelin value begins when restaurants move from comfort and price to **culinary identity and execution**.

Differentiation comes not only from the food, but from how the **entire dining experience** is designed and delivered.

At the highest level, dining becomes an **orchestrated ritual**: food is the foundation, and the experience is the art.

TEXT-BASED REGRESSION

Model 1a — Full Feature Specification (60 variables)

$$\text{StarNumeric}_i = \beta_0 + \beta_1(\text{Feature}_{1i}) + \beta_2(\text{Feature}_{2i}) + \dots + \beta_{60}(\text{Feature}_{60i}) + \varepsilon_i$$

Model 1b — Significant Feature Specification (32 variables)

$$\text{StarNumeric}_i = \beta_0 + \beta_1(\text{Feature}_{1i}) + \beta_2(\text{Feature}_{2i}) + \dots + \beta_{32}(\text{Feature}_{32i}) + \varepsilon_i$$

Model 2a — Model 1b + Price + is_home_cuisine

$$\text{StarNumeric}_i = \beta_0 + \beta_1(\text{Feature}_{1i}) + \dots + \beta_{32}(\text{Feature}_{32i}) + \beta_{33}(\text{Price}) + \beta_{34}(\text{is_home_cuisine}) + \varepsilon_i$$

Model 2b — Model 1b + is_home_cuisine

$$\text{StarNumeric}_i = \beta_0 + \beta_1(\text{Feature}_{1i}) + \dots + \beta_{32}(\text{Feature}_{32i}) + \beta_{33}(\text{is_home_cuisine}) + \varepsilon_i$$

Model 3a — Binary Attribute Model

$$\begin{aligned} \text{StarNumeric}_i = & \beta_0 + \beta_1(\text{Food_Binary}) + \beta_2(\text{Beverage_Binary}) + \beta_3(\text{Service_Binary}) + \beta_4(\text{Menu_Binary}) + \beta_5(\text{Ambience_Binary}) \\ & + \\ & \beta_6(\text{Cleanliness_Binary}) + \beta_7(\text{Accessibility_Binary}) + \varepsilon_i \end{aligned}$$

Model 3b — Attribute Intensity Model (7 count variables)

$$\begin{aligned} \text{StarNumeric}_i = & \beta_0 + \beta_1(\text{Food_Count})_i + \beta_2(\text{Beverage_Count})_i + \beta_3(\text{Service_Count})_i + \beta_4(\text{Menu_Count})_i + \beta_5(\text{Ambience_Count})_i \\ & + \\ & \beta_6(\text{Cleanliness_Count})_i + \beta_7(\text{Accessibility_Count})_i + \varepsilon_i \end{aligned}$$

Model 4 — Attribute Model with Controls (7 binary + 20 controls)

$$\begin{aligned} \text{StarNumeric}_i = & \beta_0 + \beta_1(\text{Food_Binary})_i + \beta_2(\text{Beverage_Binary})_i + \beta_3(\text{Service_Binary})_i + \beta_4(\text{Menu_Binary})_i + \beta_5(\text{Ambience_Binary})_i + \beta_6(\text{Cleanliness_Binary})_i \\ & + \\ & \beta_7(\text{Accessibility_Binary})_i + \beta_8(\text{is_home_cuisine})_i + \beta_9(\text{GreenStar})_i + \gamma_1(\text{Country_1})_i + \dots + \gamma_{18}(\text{Country_18})_i + \varepsilon_i \end{aligned}$$

Model 5 – Interactions

$$\text{StarNumeric}_i = \beta_0 + \beta_1(\text{featX}) + \beta_2(\text{is_home_cuisine}) + \beta_3(\text{featX})(\text{is_home_cuisine}) + \varepsilon_i$$

TEXT-BASED REGRESSION

Model Fit

(DV: Michelin stars 1–3, N = 18,422)

- **Model 3a (binary attributes):** $R^2 \approx 0.04$
- **Model 3b (attribute intensity/counts):** $R^2 \approx 0.11$
- **Model 4 (binary + geography/controls):** $R^2 \approx 0.08$
- **Model 5 (feature $\times$ home-cuisine interactions):** $R^2 \approx 0.15$

Interpretation

- R^2 values are **modest but typical** for cross-sectional rating data: attributes explain a **meaningful but limited** share of variation in stars.
- Moving from **binary** $\rightarrow$ **intensity** (Model 3 $\rightarrow$ 3B) **almost triples R^2** , showing that *how many* ways a restaurant exhibits quality in a dimension matters, not just whether the dimension appears.
- Adding **geography/context** (Model 4) improves fit over Model 3, indicating that country and context are **non-trivial covariates** alongside the core attributes.
- Adding **home-cuisine interactions** (Model 5) improves fit but leaves key coefficients essentially unchanged $\rightarrow$ the core ranking (**cleanliness** $\gg$ **food/service** $\gg$ **others**) is robust, with no evidence of a systematic “home-cuisine” adjustment.

TEXT-BASED REGRESSION

Which broad dimensions matter most for Michelin Stars, and why?

- **Cleanliness behaves as a threshold (Models 3a, 3b, 4).**

Cleanliness/Hygiene has a +2.56 (Model 3a) to +2.63 (Model 4) effect on a 0–3 scale ($\approx 10\text{--}12\times$ any other attribute), and Cleanliness_Count = +2.23 (Model 3b) with effectively only 0/1 values—so when “overall sanitation consistency” is mentioned, hygiene behaves like a minimum standard and stars beyond that are driven by other dimensions. However, this feature appears in only 51 restaurants out of $\sim 18k$, so the effect is estimated on a small, possibly non-representative subset where inspectors likely flag exceptionally clean restaurants while assuming a baseline hygiene level for most.

- **Food and service are the main continuous levers (Model 3b).**

In the intensity model, Food = +0.135 and Service = +0.102 per feature, combining to about +1.45 stars at maximum observed counts → once hygiene is acceptable, higher ratings go to restaurants with many distinct food and service strengths, not just one.

- **Menu and beverage are positive but secondary (Models 3a, 3b).**

Menu and Beverage have smaller effects (+0.04, +0.07 in Model 3a; +0.056, +0.036 per feature in Model 3b) → they fine-tune outcomes among already strong restaurants rather than substitute for food and service quality.

- **Ambience and accessibility are weakly penalized (Models 3a, 3b).**

Ambience and Accessibility are negative (−0.06, −0.09 in Model 3a; −0.046, −0.041 per feature in Model 3b) → when the write-up leans heavily on décor or convenience, there is relatively less emphasis on technical execution, consistent with a guide that prioritizes craft over vibe and convenience.

→ To improve Michelin prospects, restaurants should first meet a stringent cleanliness standard, then systematically strengthen food and service across multiple features (e.g., food presentation/plating, team coordination, service professionalism), while treating menu structure, beverage programs, ambience, and accessibility as secondary design choices rather than primary drivers of star ratings.

LIMITATIONS & NEXT STEPS

- **Inspector-written text, not customer sentiment**

All text features come from Michelin inspector descriptions, not customer reviews or behavioral data → we observe how inspectors describe restaurants, not demand, satisfaction, or willingness to pay.

- **Selected and cross-sectional sample**

The dataset includes only restaurants in the Guide (0-3 stars / Bib) and one time period (2024 results) → results describe relative differences within Michelin-listed restaurants, not why some restaurants are excluded or how ratings evolve over time.

- **Modeling choices and omitted variables**

We use linear OLS on an ordinal 0-3 star outcome with modest R^2 and no controls for factors like chef reputation, age of restaurant, local competition, or inspector heterogeneity → coefficients are correlational, not causal, and a large fraction of variation remains unexplained.

RECOMMENDATIONS

Michelin stars reward technical mastery and cohesive dining identity — not ambience or affordability. Restaurants should amplify food and service narratives, while using ambience and value as enhancers rather than anchors.

1. Food: strongest and most consistent driver of stars

- Sharpen a clear culinary concept and execute it with very high consistency.
- Invest in a few signature dishes that are both technically impressive and memorable, since that type of feature is what shows up most strongly in your model.

2. Service quality & team coordination: clear positive association

- Systematize service choreography (greeting, pacing, explanations, clearing plates) so the experience feels smooth and intentional.
- Train staff to be able to explain dishes, ingredients, and pairings confidently, since “expertise” shows up as a positive feature in the data.

3. Ambience & environment: supportive but secondary

- Ensure ambience supports the culinary concept (lighting, music, spacing, tableware).
- Remove obvious friction points (cramped seating, noisy/echoey rooms, awkward layout) that could generate negative ambience mentions.

4. Leverage the “second priority dimension” to differentiate market positioning

- Food quality is consistently emphasized across all contexts, so differentiation relies on the second-priority dimension — what makes the experience unique beyond food.
- That dimension varies by region and cuisine, so restaurants should align their narrative (ambience, menu creativity, or service storytelling) with the cultural expectations of their specific market.

THANK YOU

bon appétit!



APPENDIX - API Prompt

You are an expert in marketing research and attribute/feature extraction.

Your job is to analyze ONE customer review and detect ONLY the attributes and features from the predefined taxonomy that are EXPLICITLY mentioned in the review text.

STRICT EXTRACTION MODE:

1. SOURCE OF TRUTH

- You MUST use ONLY:
 - the review text
 - the taxonomy below
- You MUST NOT use:
 - world knowledge about restaurants
 - assumptions about typical service
 - your own expectations or logic

2. WHAT COUNTS AS A MENTION

- A mention MUST be supported by EXPLICIT textual evidence:
 - direct wording in the review, OR
 - very close paraphrase with the SAME concrete meaning.
- If you cannot point to at least ONE short quote from the review that supports the attribute or feature, you MUST NOT include it.
- Vague or generic praise/complaint without a clear link to a specific

attribute/feature in the taxonomy MUST be ignored.

3. WHAT DOES *NOT* COUNT (VERY IMPORTANT)

- You MUST NOT “connect the dots” or infer beyond the text.
- You MUST NOT include attributes or features based on:
 - implications
 - what probably happened
 - typical restaurant behavior
 - how things usually work

Examples of INVALID inferences:

- “Ask for a table by the window” ≠ “Reservation & booking handling”
- “The restaurant was busy” ≠ “Reservation systems, booking ease & seating availability”
- “Fancy place” ≠ any specific feature about service, menu, or beverages
- “The food was tasty” ≠ “Ingredient freshness, quality & sourcing” unless quality/sourcing is explicit
- “We got a great view” ≠ “Reservation systems” or “queue management”

4. MATCHING CRITERIA

- ATTRIBUTE:
 - An attribute is considered mentioned if at least ONE of its features is explicitly mentioned in the review.
- FEATURE:
 - A feature is considered mentioned ONLY if the review clearly talks about that specific operational aspect.
 - Do NOT guess or stretch meanings just to assign features.

APPENDIX - API Prompt

5. OUTPUT FORMAT

You MUST return ONLY a single JSON object in this exact format:

```
{  
  "attributes": [  
    "Exact Attribute Name from TAXONOMY"  
  ],  
  "features": [  
    "Exact Feature Name from TAXONOMY"  
  ]  
}
```

- Every attribute name MUST exactly match one of the attribute labels in the taxonomy.
- Every feature name MUST exactly match one of the feature labels in the taxonomy.
- Do NOT return duplicates.
- Do NOT return anything else:
 - No comments
 - No explanations
 - No reasoning
 - No sentiment
 - No invented labels

TAXONOMY

<<TAXONOMY>>

REVIEW TEXT:

"<<REVIEW_TEXT>>"

APPENDIX - Attributes & Features

1. FOOD QUALITY (Freshness, Flavor & Culinary Execution)

- Ingredient freshness, quality & sourcing
- Seasonal, local & market-driven ingredients
- Taste, flavor balance & seasoning
- Culinary technique, skill & execution
- Creativity vs. traditional preparation
- Dish variety, protein options & specialty items
- House-made / artisanal components
- Signature dishes & specialties
- Presentation & plating quality
- Portion size & consistency

2. BEVERAGE QUALITY (Taste, Craft & Selection)

- Wine list breadth, curation & recommendations
- Cocktail craftsmanship & mixology quality
- Beer, spirits & specialty beverage offerings
- Premium tea, coffee & non-alcoholic drink selections
- Natural/organic/regional beverage options
- Beverage taste, flavor quality & consistency
- Beverage pairings (wine/sake/beer)

3. SERVICE QUALITY (Interaction, Expertise & Efficiency)

- Staff friendliness, warmth & hospitality
- Professionalism, expertise & explanations
- Personalized, attentive & accommodating service
- Service efficiency, pacing & organization
- Reservation & booking handling
- Order accuracy & smooth flow
- Chef / owner interaction
- Team coordination & professionalism
- Multilingual or culturally aware communication

4. MENU VARIETY & CUSTOMIZATION (Choice, Flexibility & Innovation)

- Seasonal & rotating menus
- Menu formats (tasting menus, prix fixe, à la carte, sharing)
- Dietary accommodations (vegetarian, vegan, gluten-free, allergies)
- Customizable dishes, substitutions & flexibility
- Signature & specialty menu items
- International, fusion, regional & traditional cuisines
- Portion flexibility (half portions, tapas-style, small plates)

5. AMBIENCE & INTERIOR ENVIRONMENT (Sensory Experience & Atmosphere)

- Interior decor, aesthetic design & thematic style
- Lighting, music, noise & overall atmosphere
- Seating comfort, spacing & layout
- Scenic views, natural lighting & outdoor/terrace seating
- Open kitchen, chef's counter & visibility
- Artistic, cultural or thematic interior elements
- Cozy, intimate, relaxed atmospheres
- Vibrant, lively or entertainment-driven settings
- Garden, alfresco or nature-integrated dining environments

6. CLEANLINESS & HYGIENE (Sanitation & Maintenance)

- Dining area cleanliness & upkeep
- Kitchen & food-prep hygiene standards
- Restroom cleanliness & maintenance

- Air quality, ventilation & odor control
- Overall sanitation consistency

7. PRICE & VALUE (Affordability, Fairness & Perceived Worth)

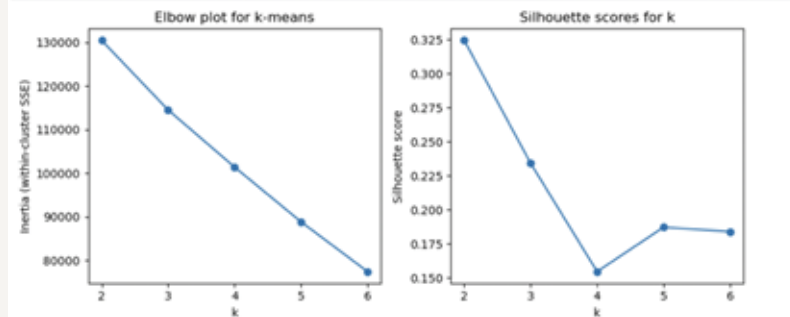
- Overall value for money
- Price level & market positioning
- Portion size relative to price
- Set menus, promotions, lunch specials & deals
- Transparent, fair & inclusive pricing
- Premium pricing for elevated experiences

8. ACCESSIBILITY & CONVENIENCE (Location, Ease & Logistics)

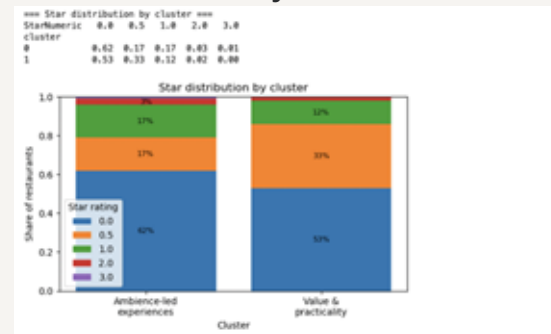
- Location convenience & transport access
- Parking availability & guest access infrastructure
- Reservation systems, booking ease & seating availability
- Wait time & queue management
- Seating configuration & indoor/outdoor flexibility
- Accessibility for mobility-impaired guests
- Take-out, delivery, catering & integrated hotel/venue access

APPENDIX - Cluster Analysis

K-Score

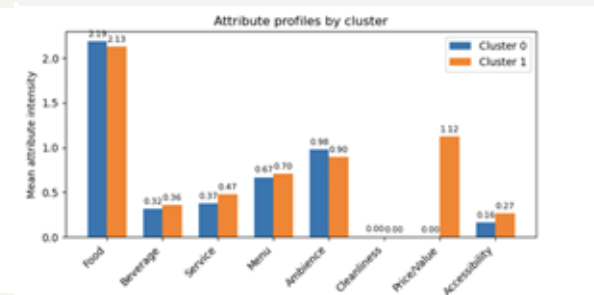


Star Distribution by cluster



Cluster Analysis

cluster	attr_FOOD_QUALITY_Freshness_Flavor_Culinary_Execution \		
0		2.188	
1		2.127	
cluster	attr_BEVERAGE_QUALITY_Taste_Craft_Selection \		
0		0.308	
1		0.356	
cluster	attr_SERVICE_QUALITY_Interaction_Expertise_Efficiency \		
0		0.371	
1		0.473	
cluster	attr_MENU_VARIETY_CUSTOMIZATION_Choice_Flexibility_Innovation \		
0		0.668	
1		0.784	
cluster	attr_AMBIENCE_INTERIOR_ENVIRONMENT_Sensory_Experience_Atmosphere \		
0		0.962	
1		0.895	
cluster	attr_CLEANLINESS_HYGIENE_Sanitation_Maintenance \		
0		0.003	
1		0.002	
cluster	attr_PRICE_VALUE_Affordability_Fairness_Perceived_Health \		
0		0.000	
1		1.124	
cluster	attr_ACCESSIBILITY_CONVENIENCE_Location_Ease_Logistics StarNumeric \		
0		0.159	0.344
1		0.267	0.325
cluster	is_home_cuisine GreenStar PriceLevel		
0		0.892	0.836
1		0.808	0.827



APPENDIX - TEXT ANALYSIS

Model 1a

```
=====
                        OLS Regression Results
=====
Dep. Variable:          StarNumeric    R-squared:                0.139
Model:                  OLS            Adj. R-squared:          0.137
Method:                 Least Squares   F-statistic:              49.57
Date:                   Mon, 01 Dec 2025 Prob (F-statistic):      0.00
Time:                   16:05:07        Log-Likelihood:          -13203.
No. Observations:      18422           AIC:                    2.653e+04
Df Residuals:          18361           BIC:                    2.701e+04
Df Model:              60
Covariance Type:       nonrobust
```

Model 2a

```
=====
                        OLS Regression Results
=====
Dep. Variable:          StarNumeric    R-squared:                0.139
Model:                  OLS            Adj. R-squared:          0.137
Method:                 Least Squares   F-statistic:              87.15
Date:                   Mon, 01 Dec 2025 Prob (F-statistic):      0.00
Time:                   16:39:03        Log-Likelihood:          -13210.
No. Observations:      18422           AIC:                    2.649e+04
Df Residuals:          18387           BIC:                    2.676e+04
Df Model:              34
Covariance Type:       nonrobust
```

Model 1b

```
=====
                        OLS Regression Results
=====
Dep. Variable:          StarNumeric    R-squared:                0.138
Model:                  OLS            Adj. R-squared:          0.137
Method:                 Least Squares   F-statistic:              92.20
Date:                   Mon, 01 Dec 2025 Prob (F-statistic):      0.00
Time:                   16:06:05        Log-Likelihood:          -13215.
No. Observations:      18422           AIC:                    2.650e+04
Df Residuals:          18389           BIC:                    2.675e+04
Df Model:              32
Covariance Type:       nonrobust
```

Model 2b

```
=====
                        OLS Regression Results
=====
Dep. Variable:          StarNumeric    R-squared:                0.139
Model:                  OLS            Adj. R-squared:          0.137
Method:                 Least Squares   F-statistic:              89.72
Date:                   Mon, 01 Dec 2025 Prob (F-statistic):      0.00
Time:                   16:47:27        Log-Likelihood:          -13211.
No. Observations:      18422           AIC:                    2.649e+04
Df Residuals:          18388           BIC:                    2.676e+04
Df Model:              33
Covariance Type:       nonrobust
```

APPENDIX - TEXT ANALYSIS

Model 3a

Model 1

OLS Regression Results

Dep. Variable:	StarNumeric	R-squared:	0.037
Model:	OLS	Adj. R-squared:	0.036
Method:	Least Squares	F-statistic:	100.2
Date:	Mon, 01 Dec 2025	Prob (F-statistic):	2.43e-144
Time:	16:48:01	Log-Likelihood:	-14242.
No. Observations:	18422	AIC:	2.850e+04
Df Residuals:	18414	BIC:	2.856e+04
Df Model:	7		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	0.1040	0.014	7.291	0.000	0.076	0.132
Food_Quality_Binary	0.2016	0.014	14.729	0.000	0.175	0.228
Beverage_Quality_Binary	0.0657	0.009	7.329	0.000	0.048	0.083
Service_Quality_Binary	0.1307	0.009	14.883	0.000	0.113	0.148
Menu_Variety_Binary	0.0443	0.008	5.648	0.000	0.029	0.060
Ambience_Interior_Binary	-0.0563	0.010	-5.527	0.000	-0.076	-0.036
Cleanliness_Hygiene_Binary	2.5637	0.524	4.889	0.000	1.536	3.592
Accessibility_Convenience_Binary	-0.0891	0.014	-6.534	0.000	-0.116	-0.062

Omnibus:	7345.302	Durbin-Watson:	0.102
Prob(Omnibus):	0.000	Jarque-Bera (JB):	32197.813
Skew:	1.946	Prob(JB):	0.00
Kurtosis:	8.177	Cond. No.	205.

Model 3b

Model 01

OLS Regression Results

=====						
Dep. Variable:	StarNumeric	R-squared:	0.111			
Model:	OLS	Adj. R-squared:	0.110			
Method:	Least Squares	F-statistic:	326.9			
Date:	Mon, 01 Dec 2025	Prob (F-statistic):	0.00			
Time:	17:02:46	Log-Likelihood:	-13507.			
No. Observations:	18422	AIC:	2.703e+04			
Df Residuals:	18414	BIC:	2.709e+04			
Df Model:	7					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]
const	0.0330	0.008	3.894	0.000	0.016	0.050
Food_Quality_Count	0.1345	0.003	38.574	0.000	0.128	0.141
Beverage_Quality_Count	0.0364	0.007	5.373	0.000	0.023	0.050
Service_Quality_Count	0.1016	0.006	16.594	0.000	0.090	0.114
Menu_Variety_Count	0.0561	0.005	12.241	0.000	0.047	0.065
Ambience_Interior_Count	-0.0459	0.009	-5.100	0.000	-0.064	-0.028
Cleanliness_Hygiene_Count	2.2256	0.504	4.416	0.000	1.238	3.214
Accessibility_Convenience_Count	-0.0407	0.011	-3.569	0.000	-0.063	-0.018
=====						
Omnibus:	7007.670	Durbin-Watson:	0.225			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	31306.151			
Skew:	1.833	Prob(JB):	0.00			
Kurtosis:	8.229	Cond. No.	325.			
=====						

APPENDIX - TEXT ANALYSIS

Model 4

OLS Regression Results						
Dep. Variable:	StarNumeric	R-squared:	0.082			
Model:	OLS	Adj. R-squared:	0.081			
Method:	Least Squares	F-statistic:	60.96			
Date:	Mon, 01 Dec 2025	Prob (F-statistic):	2.82e-316			
Time:	17:24:26	Log-Likelihood:	-13796.			
No. Observations:	18422	AIC:	2.765e+04			
Df Residuals:	18394	BIC:	2.787e+04			
Df Model:	27					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.0416	0.032	1.298	0.194	-0.021	0.104
Food_Quality_Binary	0.1818	0.013	13.490	0.000	0.155	0.208
Beverage_Quality_Binary	0.0800	0.009	8.920	0.000	0.062	0.098
Service_Quality_Binary	0.1379	0.009	15.564	0.000	0.121	0.155
Menu_Variety_Binary	0.0331	0.008	4.148	0.000	0.017	0.049
Ambience_Interior_Binary	-0.0369	0.010	-3.627	0.000	-0.057	-0.017
Cleanliness_Hygiene_Binary	2.6272	0.512	5.128	0.000	1.623	3.631
Accessibility_Convenience_Binary	-0.0786	0.014	-5.796	0.000	-0.105	-0.052
is_home_cuisine	0.0123	0.014	0.848	0.397	-0.016	0.041
GreenStar_Binary	0.4925	0.021	23.821	0.000	0.452	0.533
Country_Belgium	0.0122	0.034	0.359	0.720	-0.055	0.079
Country_China	0.1796	0.032	5.596	0.000	0.117	0.242
Country_Denmark	0.2161	0.057	3.816	0.000	0.105	0.327
Country_Europe	-0.0164	0.036	-0.456	0.648	-0.087	0.054
Country_France	0.0467	0.030	1.552	0.121	-0.012	0.106
Country_Germany	0.0627	0.032	1.989	0.047	0.001	0.124
Country_Italy	0.0115	0.031	0.376	0.707	-0.048	0.071
Country_Japan	0.2453	0.033	7.490	0.000	0.181	0.309
Country_Netherlands	0.0486	0.036	1.334	0.182	-0.023	0.120
Country_Other	-0.0226	0.031	-0.734	0.463	-0.083	0.038
Country_Portugal	0.0737	0.047	1.577	0.115	-0.018	0.165
Country_Singapore	0.1074	0.041	2.595	0.009	0.026	0.189
Country_South Korea	0.1452	0.047	3.098	0.002	0.053	0.237
Country_Spain	0.0862	0.032	2.704	0.007	0.024	0.149
Country_Switzerland	0.0690	0.036	1.931	0.053	-0.001	0.139
Country_Thailand	0.0255	0.039	0.660	0.509	-0.050	0.101
Country_US	0.0493	0.031	1.585	0.113	-0.012	0.110
Country_United Kingdom	-0.0175	0.032	-0.541	0.588	-0.081	0.046
Omnibus:	7021.013	Durbin-Watson:	0.163			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	29710.864			
Skew:	1.862	Prob(JB):	0.00			
Kurtosis:	7.985	Cond. No.	209.			

APPENDIX - TEXT ANALYSIS

Model 4

OLS Regression Results						
Dep. Variable:	StarNumeric	R-squared:	0.082			
Model:	OLS	Adj. R-squared:	0.081			
Method:	Least Squares	F-statistic:	60.96			
Date:	Mon, 01 Dec 2025	Prob (F-statistic):	2.82e-316			
Time:	17:24:26	Log-Likelihood:	-13796.			
No. Observations:	18422	AIC:	2.765e+04			
Df Residuals:	18394	BIC:	2.787e+04			
Df Model:	27					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.0416	0.032	1.298	0.194	-0.021	0.104
Food_Quality_Binary	0.1818	0.013	13.490	0.000	0.155	0.208
Beverage_Quality_Binary	0.0800	0.009	8.920	0.000	0.062	0.098
Service_Quality_Binary	0.1379	0.009	15.564	0.000	0.121	0.155
Menu_Variety_Binary	0.0331	0.008	4.148	0.000	0.017	0.049
Ambience_Interior_Binary	-0.0369	0.010	-3.627	0.000	-0.057	-0.017
Cleanliness_Hygiene_Binary	2.6272	0.512	5.128	0.000	1.623	3.631
Accessibility_Convenience_Binary	-0.0786	0.014	-5.796	0.000	-0.105	-0.052
is_home_cuisine	0.0123	0.014	0.848	0.397	-0.016	0.041
GreenStar_Binary	0.4925	0.021	23.821	0.000	0.452	0.533
Country_Belgium	0.0122	0.034	0.359	0.720	-0.055	0.079
Country_China	0.1796	0.032	5.596	0.000	0.117	0.242
Country_Denmark	0.2161	0.057	3.816	0.000	0.105	0.327
Country_Europe	-0.0164	0.036	-0.456	0.648	-0.087	0.054
Country_France	0.0467	0.030	1.552	0.121	-0.012	0.106
Country_Germany	0.0627	0.032	1.989	0.047	0.001	0.124
Country_Italy	0.0115	0.031	0.376	0.707	-0.048	0.071
Country_Japan	0.2453	0.033	7.490	0.000	0.181	0.309
Country_Netherlands	0.0486	0.036	1.334	0.182	-0.023	0.120
Country_Other	-0.0226	0.031	-0.734	0.463	-0.083	0.038
Country_Portugal	0.0737	0.047	1.577	0.115	-0.018	0.165
Country_Singapore	0.1074	0.041	2.595	0.009	0.026	0.189
Country_South Korea	0.1452	0.047	3.098	0.002	0.053	0.237
Country_Spain	0.0862	0.032	2.704	0.007	0.024	0.149
Country_Switzerland	0.0690	0.036	1.931	0.053	-0.001	0.139
Country_Thailand	0.0255	0.039	0.660	0.509	-0.050	0.101
Country_US	0.0493	0.031	1.585	0.113	-0.012	0.110
Country_United Kingdom	-0.0175	0.032	-0.541	0.588	-0.081	0.046
Omnibus:	7021.013	Durbin-Watson:	0.163			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	29710.864			
Skew:	1.862	Prob(JB):	0.00			
Kurtosis:	7.985	Cond. No.	209.			

APPENDIX - TEXT ANALYSIS

Model 5

```

                        OLS Regression Results
=====
Dep. Variable:          StarNumeric    R-squared:                0.146
Model:                  OLS            Adj. R-squared:          0.140
Method:                 Least Squares   F-statistic:              26.91
Date:                   Mon, 01 Dec 2025 Prob (F-statistic):       0.00
Time:                   22:51:25        Log-Likelihood:         -13135.
No. Observations:       18422          AIC:                    2.650e+04
Df Residuals:           18305          BIC:                    2.742e+04
Df Model:               116
Covariance Type:        nonrobust

```

APPENDIX - GOOGLE DRIVE LINK

<https://drive.google.com/drive/folders/1P9UM2iPpLMjSWZ-BCuNPsp1yfUKhINB3?usp=sharing>